MIL-E-1/406C
20 February 1986
SUPERSEDING
MIL-E-1/405B
26 September 1978

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 6BA7

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Pentagrid converter

Outline - - - - - 6-3(EIA)

Base - - - - - E9-1

Envelope - - - - - T6-1/2

Cathode - - - - - Coated unipotential

Base connections:

Pin No. - - - - -	1	2	3	4	5	6	7	8	9
Element - - - - -	g2,g4	g1	k	h	h	g5,sh	g3	sh	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec2 & 4	Ec1	Ec3	Pp	Pg2	Ik	Ehk	Alt
Unit:	Vdc	Vdc	Vdc	Vdc	Vdc	W	W	mAdc	v	ft
Maximum:	6.9	330	110	---	---	2.2	1.7	24	100	(See note 1)
Minimum:	5.7	---	---	---	---	---	---	---	---	
TEST CONDITIONS:	6.3	250	100	0	-1.0	---	---	---	---	---

GENERAL:

© Qualification - Not required

© denotes changes

Source: https://assist.dla.mil -- Downloaded: 202640-15T13:19Z
Check the source to verify that this is the current version before use.

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 1	2					
1231	Emission	3	Eb = Ec1 = Ec2 & 4 = 15 Vdc	Is	50	---	mAdc
1266	Total grid current	3, 4	Ec3 = -2 Vdc	Ic3	0	-1.5	µAdc
1306	Transconductance	5	Ec3 = 0; Eb = 100 Vdc	Sm	6,200	9,800	µmhos
1326	Conversion transconductance	4		Sc	600	1,300	µmhos
1256	Electrode current (1) (anode)	4		Ib	2.7	5.1	mAdc
1256	Electrode current (screen grid)	4		Ic2 & 4	7.5	13.5	mAdc
1201	Short and discontinuity detection	6		---	---	---	---
	Quality conformance inspection, part 2						
1031	Low-frequency vibration		Rp = 10,000 ohms; Ec1 = Ec3 = -1.5 Vdc	Ep	---	300	mVac
1301	Heater current			If	275	325	mA
1336	Heater-cathode leakage			Ihk	---	20	µAdc
1256	Electrode current (2) (anode)	5	Ec3 = 0; Eb = 100 Vdc	Ib	22	44	mA
1331	Direct-interelectrode capacitance		Without shield Without shield Without shield	Cg3p Cg3-all Cp-all	--- 7.9 6.0	0.19 11.3 10.6	pF pF pF
1211	Insulation of electrodes			---	---	---	---
1105	Permanence of marking			---	---	---	---
	Quality conformance inspection, part 3						
1501	Intermittent life	7	Group A; Ec2 & 4 = 135 Vdc; Ec1 = -16 Vdc; Ehk = 100 Vdc; Ec3 = 0	---	---	---	---
---	Intermittent life-test end points (500 hours):						
1306	Transconductance	5	Ec3 = 0; Eb = 100 Vdc	Sm	4,800	---	µmhos
1326	Conversion transconductance	4		Sc	480	---	µmhos

Source: https://assist.dla.mil -- Downloaded: 202610-15T13:19Z
Check the source to verify that this is the current version before use.

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage.
2. Unless otherwise specified, the AQL (percent defective) for all tests in QCI, part 1, shall be 1.0, inspection level II.
3. This test shall be performed at the conclusion of the holding period.
4. Insert 20,000 ohms and a suitable bypass capacitor in series with grid 1. Apply a 6.8 Vac signal to grid 1, which produces an 0.350 mAdc average grid current.
5. Tie anode and grids 2 and 4 together at the socket.
6. The AQL (percent defective) shall be 0.4, inspection level II.
7. Apply 12.4 Vac to grid 1. Ec1 may be obtained by self rectification.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, AV, MI
Air Force - 99
DLA - ES

User activities:
Navy - AS, CG, MC, OS
Air Force - 11, 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5960-3358)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:19Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T13:19Z
Check the source to verify that this is the current version before use.